

Alcon: Automated Particulate Counting for Handpiece Production

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Thesis, Purpose, Objective & Approach

Our goal is to develop an automated method for particulate counting to be used in the production testing of Alcon's Active Sentry Handpieces [Alcon Professional 2024]. Through the identification and implementation of technology options for particulate counting, these methods will give consistent results, be time efficient, and be cost effective. We plan to explore, evaluate, and integrate advanced particulate counting solutions, such as HD optics and artificial intelligence, into Alcon's handpiece production process. Our goal is to modernize testing methods that are critical to the successful operation of surgical consoles used in cataract surgeries worldwide.

This project holds significant value because it addresses a critical component of eye surgery: handpiece quality and safety. The Active Sentry handpiece is a component of Alcon's Centurion eye surgery system [Alcon Professional 2024]. After handpieces are assembled, they undergo a testing procedure that concludes with a fluid flush of the device. Any particles suspended in the fluid are captured by a filter, which now must be examined to determine if machining debris remains inside the handpiece. At present, trained technicians manually evaluate particulates generated in the final device testing process. As demand for these devices grows, current testing methods have proven unsustainable and unable to scale with production

needs. By advancing knowledge in particulate counting technology, we not only aim to enhance our quality control processes but also reduce costs, increase yield, and contribute to safer, more effective surgical procedures globally.

Automating particulate counting can transform the production line by reducing manual testing inconsistencies, minimizing time delays, and ensuring higher accuracy. The integration of AI into these systems will allow for adaptive learning, where the inspection platform becomes more accurate and efficient over time. This project not only supports Alcon's mission to improve vision care but also sets a new standard for quality assurance in medical device manufacturing. The societal impact of improved surgical outcomes and increased access to reliable eye care technology is invaluable.

Our team will explore various particle analysis techniques, aiming to identify those best suited in evaluating the shape, size, and type of particulate matter within the handpiece. Current particulate analysis techniques involve various light and optical methods. An advanced microscopy technique such as Fourier transform infrared microscopy (FTIR) is used to detect and identify smaller impurities using an infrared light source with a spot size as small as 10 μm [Rqmplus 2017]. It works by capturing infrared wavelengths reflected across a sample to identify its molecular composition and structure [Scientific Reports 2017]. Another technique known as energy dispersive X-ray (EDX) is paired with a scanning electron microscope (SEM) to create another microscopy method called SEM-EDX [Advancing Materials 2019]. This method identifies qualitative elements in a sample by detecting unique X-rays from electrons transitioning between energy levels in an atom after being displaced by an electron beam [Advancing Materials 2019]. A detector collects the X-rays, which measures the signal and interprets its composition using a specialized software [Advancing Materials 2019]. Some spectroscopy methods include Raman spectroscopy or energy dispersive spectroscopy (EDS) to attempt to identify the chemical or elemental composition respectively [Particle Technology Labs 2023]. Raman spectroscopy works by analyzing the inelastic scattering of light known as

the Raman effect when it interacts with a material's molecular vibrations or crystal lattice. The effect produces signals that can be used to describe the molecular bonds and crystal structure [Commun. Mater. 2023]. White-light interferometry is another important technique used in particulate analysis, which works by using a broadband light source and an interferometer to create an interference pattern when light reflects off the surfaces of particles [Pavliček and Mikeska 2023]. The variations in light refractions caused by differences in surface height allows for the 3D mapping of the particle's topography, which allows for the precise measurement of the dimensions and roughness [Pavliček and Mikeska 2023].

Once we identify an effective imaging technology, our next step will be to develop a semi-automatic data analysis process to accurately count and evaluate particulates. This system will upload imaging data to a MATLAB script that counts and categorizes particles based on relevant traits, such as size, density, and shape. With each iteration of the system, we will implement the design within the assembly line to check for any needed changes or errors. The testing of the system will be done as a group as each individual can spot specific errors within their role. Any flaws detected will go back to the prototype and revision team for corrections and alterations, and the new iteration of the system will be tested until a functional model is achieved.

Responsibility

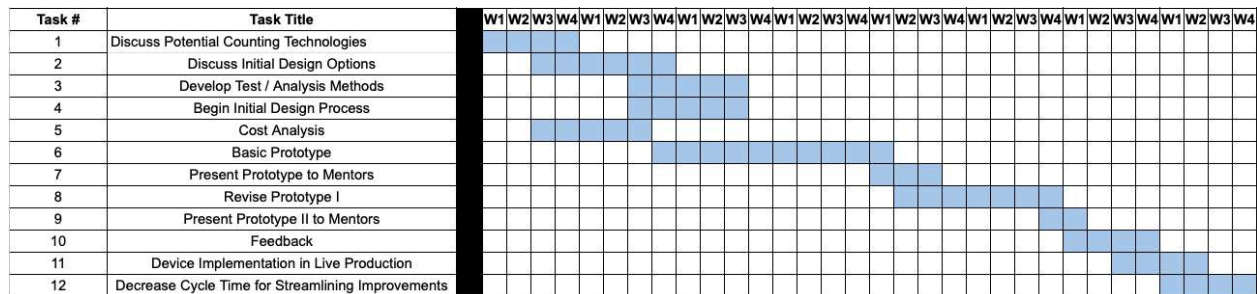
Our team is made up of six undergraduate students and two engineering mentors. The main role of the undergraduates will be in the design and testing of the proposed device. Specifically, Isaac will be responsible for leading the team – ensuring that deadlines are met, progress remains on track and on time, and maintaining communication with the engineering mentors Tom Schacht and Diego Cueto. Brian's role will be to advise the team in conducting business in conjunction with Alcon and providing insight into utilization of the device. Matthew's

responsibilities will include working with hardware and manufacturing – making sure that production of the device gets off the ground. Yolanda's and Tandel's main responsibilities are in prototype development and revision. They will be heading development of the device, confirming that changes made are effective and well documented. Landon's role will be in test development – verifying design choices and checking that said design choices align with the intentions for the device.

The engineering mentors on this project are Tom Schacht and Diego Cueto. As a Senior Manager of Manufacturing at Alcon, Tom Schacht will be a major point of contact between the team and Alcon and will be guiding the undergraduate team through the design process and specifications. Diego Cueto, Head of Manufacturing at Alcon, will be overseeing the entire project from a higher standpoint, confirming that design and development is headed in the right direction.

Timeline

+ Alcon Gantt Chart



References

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